METRIC
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COMMERCIAL ITEM DESCRIPTION

SPRAY SHIELDING TAPE, ALUMINUM COATED GLASS CLOTH, PRESSURE-SENSITIVE, FOR MECHANICAL JOINTS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for fire-resistant, pressure-sensitive adhesive, aluminized glass cloth tape for spray shielding flanged piping joints and fittings. Spray shielding tape prevents joint leakage of flammable fluids (e.g., fuel, aviation fuel, lubricating and hydraulic fluid) and non-flammable fluids of all types from impinging on adjacent surfaces, preventing fire and personnel injury.

2. CLASSIFICATION. Tape shall be available in rolls of the following width sizes: 35, 50, 100, 140, 250, and 500 millimeters (mm) (1.4, 2, 4, 5.5, 10, and 20 inches [in]). When specified (see 7.3), tape shall be available in rolls with width size of 1,000 millimeters (40 inches). Unless otherwise specified (see 7.3), the length of a roll of tape shall be 10 meters (33 feet) per roll.

3. SALIENT CHARACTERISTICS.

3.1 Material. The spray shielding tape shall be made from aluminized glass cloth material, with additional materials used as needed to meet the requirements herein. The spray shielding tape shall not contain asbestos.

3.2 Adhesive. The adhesive used for the spray shielding tape shall be a pressure-sensitive silicone or acrylic polymer applied uniformly, free of voids and lumps, to the backing of the tape and shall not require heat treatment or any other means of preparation prior to application.

3.3 Physical properties. The physical properties of the spray shielding tape shall be in accordance with table I. The spray shielding tape shall have the physical properties described in table I when tested in accordance with the tests stated herein (see 3.3.1 through 3.3.6). All testing shall be conducted as stated herein or in accordance with the cited reference. Tests may be conducted sequentially or concurrently. Tests shall be conducted on tape of the 50-millimeter and 250-millimeter width sizes.

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FSC 9390

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TABLE I. Physical properties of spray shielding tape.

Physical property	Requirement
Operating temperature range of applied surface	-4 to 150 °C (25 to 302 °F)
Operating pressure range of fluid inside protected piping	0 to 21 bar (0 to 300 psi) Up to 50 bar (725 psi), when specified (see 7.3)
Tensile strength in accordance with ASTM D882	0.7 mPa (100 psi)
Adhesion to steel in accordance with ASTM D3330/D3330M	0.5 N/m (4.2 lbf/in)
Adhesion at elevated temperature in accordance with ASTM D3654/D3654M	No slippage when installed in accordance with manufacturer's instructions
Surface flammability in accordance with International Maritime Organization (IMO) Fire Test Procedures (FTP) Code	Compliant with IMO FTP Code Part 5 when installed in accordance with manufacturer's instructions
Galvanic corrosion of steel, copper, and aluminum	No visible signs of corrosion allowed
NOTE: psi = pounds per square inch, mPa = megapascals, N/m = newton per meter, and lbf/in = pound force per inch.	

3.3.1 Environmental test. An environmental test shall be conducted to demonstrate the capability of the spray shielding tape to withstand the required operating temperatures and pressures of the shielded fluid. This test shall be conducted by installing the spray shielding tape of appropriate size on a flanged piping joint, so that the flange bolt heads are covered. The test fluid shall be an oil with a density between 700 and 1,000 kilograms per cubic meter. The test shall be conducted at the operating temperature and pressure extremes of table I. The test shall be conducted while at the highest pressure, and the spray shielding tape shall be tested at the highest and lowest temperature extremes for a period of at least 1 hour each. Prior to testing, the flange joint under test shall have been prepared with an intentional leak of at least 1 percent of its diameter. The spray shielding tape shall not burst, rip, tear, deform, or become loose from the flange joint surfaces such that pressurized fluid is allowed to spray away from the flange joint during the duration of the test. Test fluid that forms droplets and escapes from the spray shielding tape but does not spray away from the flange joint is acceptable.

3.3.2 Tensile strength test. A tensile strength test shall be conducted in accordance with ASTM D882. Failure of the spray shielding tape to meet the requirements of table I shall be cause for rejection.

3.3.3 Adhesion to steel (peel) test. An adhesion to steel test shall be conducted in accordance with ASTM D3330/D3330M. Failure of the spray shielding tape to meet the requirements of table I shall be cause for rejection.

3.3.4 Adhesion at elevated temperature test. An adhesion at elevated temperature test shall be conducted in accordance with ASTM D3654/D3654M. Failure of the spray shielding tape to meet the requirements of table I shall be cause for rejection.

3.3.5 Flammability test. A flammability test shall be conducted in accordance with IMO FTP Code Part 5. Failure of the spray shielding tape to meet the requirements of IMO FTP Code Part 5 shall be cause for rejection.

3.3.6 Corrosion test. A corrosion test shall be performed on 150-millimeter (6-inch) samples of commonly available commercial steel, copper, and aluminum pipe. Spray shielding tape shall be evenly spiral wrapped, with an equal amount of unwrapped areas, around the sample pipes being tested. The samples shall be suspended in tap water in a sealed container for 144 hours. At the end of this exposure period, the tape shall be removed, and the metal surfaces shall be examined visually for any pitting or other signs of corrosion. If the taped metal surfaces are more corroded than the bare metal surfaces, then the tape shall be rejected.

3.4 Marking. Unless otherwise specified (see 7.3), each roll shall be marked in or on the edge of the core with the manufacturer's name, product designation, CID number, width, and date of manufacture.

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3.5 Shelf-life. When packaged in accordance with the packaging requirement (see 6) and stored under normal conditions in a temperate warehouse, where temperatures between -15 and 50 °C (5 and 122 °F) are possible, the tape shall meet all the requirements of this CID for a minimum of 3 years after date of receipt at destination.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The following market acceptability criteria is necessary to document the quality of the product to be provided under this CID. The company furnishing the tape must have been producing a product meeting the requirements of this CID for at least 6 months.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The width shall be expressed as a 4-digit number in millimeters (e.g., 0035, 0050, 0100, 0140, 0250, 0500).

CID Number	-	Tape Width (mm)
AA59996	-	XXXX

Example: PIN AA59996-0500 designates a 500-millimeter width spray shielding tape

7.2 Source of documents.

7.2.1 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.2.2 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/browse/index/far.

7.2.3 IMO. IMO standards are available from IMO Publishing, 4 Albert Embankment, London SE1 7SR, United Kingdom or online at www.imo.org.

7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. If 1,000-millimeter width of tape is required (see 2).
c. Width of tape, length (if other than 10 meters [33 feet]) and number of rolls (see 2).
d. If pressure rating is higher than 21 bar (see table I).
e. Markings if different from herein (see 3.4).
f. Manner of compliance with product conformance provisions (see 5).
g. Packaging requirements (see 6).

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7.4 Key words.

Flange shield

Piping

Spray protection

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – MR
Navy – SH

Review Activities:
Army – AV, MI
Navy – AS, CG
DLA – CC, CQ

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
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